**<u>Topic 1: Genes and evolution</u>**
**Book: 1 The Science of Psychology (3-6), 3 Biology and Behavior (69, 106-116)**
Psychological science is the study of mind, brain and behavior.
Mind refers to mental activity. Mental activity results from biological processes within the brain. Behavior describes the totality of observable human actions.

**The genetic basis**
Does a violent person have violent genes?

- In the seventies: no
  Believed to be caused by the environment.
- In the late nineties: yes
  genes cause the behavior.
- Today: Behaviour is believed to be based on the interaction between genes and environment.

The activities of cells are the basis of psychological processes. However, biology alone does not explain our behavior. since aspects of our biology interact with our environments.

A cell has a nucleus, in the nucleus there are 23 pairs of chromosomes so 46 chromosomes in total.
In each chromosome there is a single molecule of DNA with a double helix structure.
Each DNA molecule consists of two strands that wind around each other. Each strand has a backbone made of phosphate and sugar. Attached to each sugar is one of the following four:

- **A**denine
- **T**hymine
- **G**uanine
- **C**ytosine

Adenine forms a base pair with thymine and Guanine with Cytosine.
These (ATGC) contain biological information, these meaningful parts are the genes.

Genes: the units of heredity(erfelijkheid) that help determine the characteristics of an organism.
Gene expression: whether a particular gene is turned on or off.
Chromosomes: structures within the cell body that are made up of DNA, segments of which comprise(omvatten) individual genes.

Genes provides information on how to make a protein. Those are made on ribosomes in the cells. This only happens if the gene actually gets expressed (on/off). All cells have the same genes, they just express different parts of their genes. They can be turned off by metalation (metale groups can be attached to a gene which results in a gene turned off).
This is controlled by the biochemical environment inside the cell. The biochemical environment inside the cell is influenced by the environment outside the cell, timing in development, the overall environment, experience, behavior.

Genome: an organism's complete set of DNA (all their genes).
The gnome is the master blueprint that provides detailed instructions of everything. The gnome provides the options, and the environment determines which option is taken.

Genotype: an organism's specific set of genes. It is determined at the moment of conception and never changes.
Phenotype: the overt traits and behaviors of an organism. Phenotype is determined by genotype and environment and is always changing.

You get 23 chromosome pairs from your mom and 23 from your dad. Each gene is paired with one of the others from another pair.
Allele: one specific variant of a gene. Heterozygote (alleles on locus are different) and homozygote (alleles on locus are the same) come from this.
Dominant and recessive come into play: if an allele is dominant, it will express when they are heterozygous. If an alle is recessive, it will only express when they are homozygous.
So, a dominant gene from either parent is expressed whenever it's present. A recessive gene is expressed only when it is matched with a similar gene from the other parent.

Heredity: transmission of characteristics from parents to offspring by means of genes.
Heritability: refers to the proportion of the variation in some specific trait in a population that is due to genetics.

Trait or behaviour is based on a combination of the environment of the past and present, but also gene and polygenic inheritance. This polygenic inheritance goes on for a lot of generations.
Polygenic: the trait is influenced by many genes as well as by environment.

**Evolution by natural selection**
The genome is shaped by evolution over the years: Darwin's evolution theory.
Darwin said: Small set of ancestors evolved into a bunch of different kinds, through time/evolution.
The key mechanism of evolution is natural selection. Natural selection occurs when:
Variation is among us, certain traits lead to higher reproduction rates than others, the trait is able to be given to kids/being able to get from ancestors.
What matters is the survival of genes, not the survival of individuals.

A zygote (a fertilized cell) grows through cell division. First the chromosomes duplicate. Then the cell divides into two new cells with an identical chromosome structure.
Sometimes Errors occur during cell division, leading to mutations or alterations in the DNA.
Mutations can be adaptive or maladaptive.

Recessive genes do not interfere with most people's health. For this reason the recessive genes for diseases can survive the gene pool.
Most dominant gene disorders are lethal for most of their carriers and therefore do not last in the gene pool.

The evidence for modern evolutionary theory comes from many sources.

- The fossil record: very old fossils prove that there has been enough time to evolve.
- Transitions and differences between fossils and current species.
- Resemblance between genomes.
- Humans have 23 chromosome pairs, ape 24. This can be explained by pair 2, where the center resembles the part that is the same as the apes' two different

chromosomes (kutzin). Basically, chromosome 2 might have been a fusion in mens evolution.
- Pseudogenes: Inactive gene that does not function currently but probably used to function in the bodies of ancestors.
- Distribution of species across the world.
  Continental and oceanic islands: Japan used to be connected, not anymore. Hawaii never was connected. Japan has a set of species that have a lot of variety. Hawaii didn't have those, since they were never there (or couldn't survive there).

Many People still remain skeptical about the theory of evolution.

Evolution doesn't necessarily mean improvement. It just adapts organisms to the environment. If the environment changes quickly, newly obtained features could be harmful.

**Genes and behavior**
nature vs nurture. Nowadays it has become irrelevant: it depends on each other and both have an effect.
The MAOA gene can be measured and compared to see aggressive behaviors. Studies showed that people with a low-MAOA gene who were maltreated had a higher change of aggressive behavior than people with a high-MAOA gene.

conclusion: Does a violent person have violent genes? Yes, but not completely, it only raises the probability and also still depends on the environment.
Biology and environment mutually influence each other.

**<u>Topic 2: The brain and the nervous system</u>**
**Book: Biology and Behavior (69-101)**

**Building blocks of the nervous system**
The nervous system is made up of two basic kinds of cells:
- Glia: have primarily a supportive function
- Neurons: cells that receive, integrate and transmit information in the nervous system. They operate through electrical impulses, communicate with other neurons through chemical signals and form neural networks (these networks develop through genetic influence, maturation, experience and repeated firing).

In a neuron the cell body(The site in the neuron where information from thousands of other neurons is collected and integrated.) contains the nucleus which contains the DNA. A neuron also contains an axon and dendrites. The dendrites in a neuron receive the information and signals from other neurons. The axon is a long, narrow outgrowth of a neuron and transmits the signals and sends it from the cell body to the terminal buttons (to other neutrons).

different types of neurons
- sensory receptors: is capable of translating a physical stimulus into an electric signal.
- sensory (afferent) neurons: carries information into the central nervous system
- motor (efferent) neurons: caries information out of the central nervous system
- interneurons: make connections within the nervous system(between the afferent and efferent neurons)

Sensory and motor neurons work together to control movement.
Somatosensory: the sensory nerves that provide information from the skin and muscles.

**Communication among neurons**
During the reception phase, neurons take in the chemical signals from neighboring neurons. During integration, incoming signals are assessed. And during transmission, neurons pass their own signals to yet other receiving neurons.

The neuron is coffered by a membrane. The membrane is semipermeable, which implies that only some substances can move in and out. When the neuron is at rest there are different numbers of negatively and positively charged molecules inside and outside the cell. The resting potential difference is -70 mV. The inside is negatively charged relative to the outside.
When the membrane is stimulated the potential difference can change and can even change such that the neuron fires. The potential difference will briefly be +40 mV,the action potential, so for this moment the inside of the neuron is positive relative to the outside.
The action potential travels in time along the neuron, this is known as propagation.

Achieving the positive charge on the inside (when the neuron fires) is done by using the ion channels in the membrane. $Na^+$ gets in and $k^+$ gets out, so that the inside becomes relative for a second (the moment that $Na^+$ is on the inside, but the $k^+$ is not on the outside yet).
The positive difference triggers the next sodium channels. Sothe process repeats and lets the potential travel through the axon.

Action potential is only generated when the potential difference is above -55mV, this is the excitation threshold. During the refractory period, the neuron does not fire another action potential (repolarization).

All-or-none law: Neurons either fire or do not fire. So there either is an action potential or there is not.
intensity variations are coded by: variations in the number of neurons firing and the variations in firing rate.

Neurons interact via synapses, they do this through chemical substances(neurotransmitters).
Synapse: the place where a signal passes from one nerve cell to another.
When a signal, an action potential, arrives at the presynaptic neuron a chemical substance is released by the terminal buttons. These chemical substances travel to the postsynaptic neuron through the synapse. The postsynaptic neuron then might open its channels for na+, continuing the same process and giving the signal through.
Each neuron is influenced by multiple neurons, meaning that the chance of giving an action potential is higher if more neurons fire at the next neuron.

Lock-and-key model: each receptor can be influenced by only one type of neurotransmitter.

After being released a neurotransmitter does not stay there forever, so the effect gets terminated:

- Autoreception: after a large amount of neurotransmitters are released these receptors signal that the release needs to be stopped.

- Synaptic reuptake: Neurotransmitters are used multiple times and get retaken into a presynaptic neuron.
- Enzyme deactivation: they deactivate them in the synaps.

All neurotransmitters either have an excitatory or an inhibitory effect, depending on the receptor.

Drugs influence the nervous system. They either enhance the actions of specific transmitters.

- Agonists (Enhances the actions of neurotransmitters. When they mimic neurotransmitters and bind with their receptors as if they were the real thing.)
    - Increase of precursor
    - Counteracting the cleanup enzymes
    - Blocking the re-uptake
    - Mimicking the transmitter's action
- Antagonists (impedes the actions of neurotransmitters)
    - Decrease precursor
    - Increase effectiveness cleanup enzymes
    - Enhance the re-uptake
    - Blocking of receptors

These can also be used in a positive way (increase of serotonin to help with depression for example).

**Communication of the brain with the body**

The brain is best viewed as a collection of interacting neural circuits.

The nervous system has two major parts, a central nervous system (brain and spinal cord) and a peripheral nervous system (somatic nervous system and autonomic nervous system).

Somatic nervous system: Skin/muscles/joints send signals via nerves to the spinal cord and brain(central nervous system) and the other way around.

Autonomic nervous system: receives information from internal organs and sends them back. When sending them back, you have the sympathetic nervous system: important for preparing the body for action, and the parasympathetic nervous system: for returning the body to a resting state.

The endocrine system: consists of a number of endocrine glands(klieren). Those glands release hormones that travel through the body via the bloodstream to influence thoughts, behaviors and actions.
This kind of signaling is slow, but the effects are long(er) lasting.

**Studying the brain**

19th century: Phrenology: bumps on the skull were interpreted in terms of certain personality traits. Although they were wrong it helped spread the seemingly scientific principle that the brain functions were localized.

Broca's area: a small portion of the left frontal region of the brain, crucial for the production of language.

Methods for studying the brain:

- Clinical observation of patients with brain damage.
- Experimental techniques: Manipulate an independent variable in a controlled experiment in order to inspect its effects on behavior.
    - invasive: animal studies. Requires entering the body and most of the time damaging the body.
    - TMS (Transcranial Magnetic Stimulation): a dysfunction is temporarily created by a series of magnetic impulses.
- Electrophysiology: measures electrical activity.
    - EEG: involves the measurements of electrical currents in the brain by using a bunch of electrodes.
- Brain imaging
    - MRI scan: gives information on the structure of your brain.
      fMRI: also includes the measurement of the blood flow and oxygen.
    - PET scan: gives an indication of activity when doing certain tasks, using radioactive glucose.

**The brain**

The spinal cord is seen to be composed of two distinct tissue types. The gray matter, which is dominated by neurons' cell bodies, and the white matter, which consists mostly of axons that travel between brain regions.

Brain stem: extension of the cord: controls parts that are important for survival. It also contains a network of neurons known collectively as the reticular formation and it affects general alertness.
Cerebellum: essential for coordinated movement and balance.
Cerebral cortex: outer layer
Subcortical: usually part of the limbic system, which influences emotions, motivation, memory and sensing.
Thalamus: the gateway to the cortex. It receives almost all incoming sensory information before that information reaches the cortex.
Hypothalamus: a brain structure that is involved in the regulation of bodily functions.
Hippocampus: a brain structure that is associated with the formation of memories.
Amygdala: a brain structure that serves a vital role in learning to associate things with emotional responses and in processing emotional information.
Basal ganglia: a system of subcortical structures crucial for planning and producing movement.

80% of the brain consists of cortex. The cerebral cortex is the outer layer of the cerebral hemispheres. Each hemisphere has its own cortex. The corpus callosum is a massive bridge of millions of axons and connects the hemispheres and allows information to flow between them.
Each cerebral hemisphere has four lobes:

- Occipital lobe: important for vision
- Parietal lobe: important for touch and for attention to the environment

- Temporal lobe: important for hearing, memory and language
- Frontal lobe: important for thinking, planning and movement

prefrontal cortex: the foremost portion of the frontal lobes. Important for attention, working memory, decision making,appropriate social behavior and personality.

The brain is divided into two halves which are connected by the corpus callosum, a bundle of fibers carrying information back and forth between the two halves.
Because of this many things in the brain come in pairs, the left and right part of the brain are for a big part symmetrical. So this has some exceptions, where certain sides dominate certain parts.

Split brain: a condition that occurs when the corpus callosum is surgically cat and the two hemispheres of the brain are almost completely isolated from each other.
The hemispheres now function as completely independent entities, each half has its own perceptions, thoughts and consciousness.

**<u>Topic 3: Consciousness</u>**
**Book: 4 Consciousness (121-158)**
Consciousness: one's subjective experience of the world, resulting from brain activity. It involves your sensations, thoughts, memories and anything else you are experiencing in the moment.
Consciousness arises as a function of which brain circuits are active (global workspace model).

In the 70th century they argued that the mind is physically distinct from the brain, this is called dualism. But nowadays they believe the brain and the mind are considered inseparable.
Everyone experiences consciousness personally. So we cannot know if two people experience the world in exactly the same way.

The global workspace model posits that consciousness arises as a function of which brain circuits are active.
Sometimes unconscious processes lead people to do things that their conscious minds struggle to explain, this is known as after-the-fact-explanations.

**Variations in consciousness**
Variations in "normal" consciousness: Attention, sleep and drugs.

A coma allows the brain to rest. Because people in coma do not respond to external stimuli, it is hard to know whether they have consciousness. But a few studies showed that some people in comas are conscious, this is referred to as a minimally conscious state.

Unresponsive Wakefulness Syndrome: when people have emerged from a coma, but do not respond to external stimuli for more than a month.
Brain death: the irreversible loss of brain function, no activity is found in any region of the brain.
Locked-in syndrome: all or nearly all of a person's voluntary muscles are paralyzed. They

see all the sights around them and hear every noise, but they cannot respond physically to these sights and noises.

Altering consciousness: Hypnosis, meditation and immersion in action.

Hypnosis: a social interaction during which a person, responding to suggestions, experiences changes in memory, perception and/or voluntary action.
Sociocognitive theory of hypnosis: according to this theory hypnotized people behave as they expect hypnotised people to behave.
Neodissociation theory of hypnosis: acknowledges the importance of social context to hypnosis, but it views the hypnotic state as an altered state. According to this theory, hypnosis is a trancelike state in which conscious awareness is separated or dissociated from other aspects of consciousness.

Meditation: a mental procedure that focuses attention on an external object or on a sense of awareness.
Concentrative meditation: you focus attention on one thing.
Mindfulness meditation: you let your thoughts flow freely, paying attention to them but trying not to react to them.

**Attention**
THe mind is a continuous stream and thoughts float on that stream. However, there is a limit to how many things the mind can be conscious of at the same time.
Attention refers to the process that enables you to focus selectively on some things and avoid focusing on others.

Subliminal perception: the processing of information by your senses without conscious awareness.
change blindness: because we cannot attend to everything in the vast array of visual information available we often fail to notice large changes in your environment.

Just because people fail to attend to something in the environment does not mean that they are unaffected by it.
Freudian slip: when an unconscious thought is suddenly expressed at an inappropriate time or in an inappropriate social context.
Psychologists agree that unconscious processes influence people’s thoughts and actions as they go through their daily lives.

**Sleep**
Our conscious state changes when we fall asleep.We cycle through the different stages about five times a night, one cycle takes about 90 minutes. These patterns are known as circadian rhythms.

Changes in light are registered in the suprachiasmatic nucleus. In response to this registration this region signals to the pineal gland. The pineal gland then secretes melatonin, a hormone that affects various receptors in the body, including the brain. Melatonin is necessary for circadian cycles that regulate sleep.

Stages of sleep:

- Stage 1: theta waves, short bursts of irregular waves. In this light sleep you might see fantastical images or geometric shapes. You might have the sensation of falling or that your limbs are jerking.
- Stage 2: still theta waves but also occasional bursts of activity called sleep spindles and large waves called K-complexes. Your breathing becomes more regular and you become less sensitive to external stimulation.
- Stage 3 and 4: now seen as one stage because the brain activity is nearly identical. Delta waves, large, regular brain patterns. It is often referred to as slow-wave sleep. People still process some information, because the mind continues to evaluate the environment for potential danger.
- REM sleep: beta waves, represents an awake, alert mind. There are rapid eye movements. It is sometimes called paradoxical sleep because of the paradox of a sleeping body with an awake mind. While the brain is active most of the body's muscles are paralyzed.

Sufficient sleep is important for memory and good health and is often affected by psychological disorders.

It might be that your genes influence the amount of sleep you need, as researchers have identified a gene that influences sleep. This gene (sleepless) regulates a protein that reduces action potentials in the brain. Loss of this protein leads to an 80% reduction in sleep.

Why do we sleep?

- Restorative theory: sleep allows the body, including the brain, to rest and repair itselves.
- Circadian rhythm theory: sleep has evolved to keep animals quiet during the night, when it is the most dangerous (during the dark). So humans evolved to sleep at night because our ancestors were more at risk during the night,
- Facilitation of learning theory: sleep strengthens neural connections needed for learning.

Dreams: products of an altered state of consciousness in which images and fantasies are confused with reality.

- Non-REM dreams: relatively dull
- REM dreams: typically bizarre and intense (During REM sleep the brain regions that are active are associated with motivation, emotion and reward while the region that is associated with logic is inactive.)

Activation-synthesis theory: Random brain activity occurs during sleep and this neural firing can activate mechanisms that normally interpret sensory input. The sleeping mind tries to make sense of the resulting sensory activity by synthesizing it with stored memories. From this perspective, dreams are the side effects of mental processes produced by random neural firing.

Insomnia: a sleep disorder in which people's mental health and ability to function in daily life are compromised by difficulty both falling asleep and staying asleep.

Obstructive sleep apnea: a disorder in which a person, while asleep, stops breathing because their throat closes. This results in frequent awakenings during the night.
Narcolepsy: a sleep disorder in which people experience excessive sleepiness during normal waking hours, sometimes going limp and collapsing.

**Drugs**
Addiction: drug use that remains compulsive despite its negative consequences.
Physical and psychological dependence

- Tolerance: increasing amounts of a drug needed to achieve the intended effect
- Withdrawal: physiological and psychological state characterized by feelings of anxiety, tension and cravings for the addictive substance

Stimulants: drugs that increase behavioral and mental activity and activate the central and sympathetic nervous system.
Cocaine is a triple reuptake inhibitor, it blocks the reuptake of dopamine, serotonin and ?, it also enhances its release.

Depressants: reduce behavioral and mental activity by depressing the central nervous system.
Alcohol is the most widely used depressant. It produces its affects by activating GABA receptors (the primary inhibitory neurotransmitter in the nervous system).

Opioids (narcotics): depress or slow down the central nervous system, relieve pain and suffering by binding with endorphin receptors.

Hallucinogens (psychedelics): produce alterations in cognition, mood and perception. These drugs change how users experience the world around them.
LSD creates parts of its effects during the binding with serotonin receptors (serotonin agonist).

Many commonly used drugs do not fit into these four major categories, because they produce a range of psychological effects (e.g. Marijuana, MDMA).

Amphetamines: stimulants that increase dopamine in the synapse. Their primary effect is to reduce fatigue.
Methamphetamines: breaks down into amphetamine in the body.

**<u>Topic 4: sensation and perception: vision</u>**
**Book: 5 Sensation and Perception (163-187)**
**The senses**
Sensation: the detection of physical stimuli and the transmission of this information to the brain.
Perception: the processing, organization, and interpretation of sensory signals.

Bottom-up processing: based on the physical features of the stimulus.
Top-down processing: how knowledge, expectations, or past experiences shape the interpretation of sensory information.

How you experience the world around you, brought to you through your senses, is influenced by your knowledge, expectations, and past experiences.

Transduction: the process by which sensory stimuli are converted to signals the brain can interpret.
The sensory receptors receive stimulation (physical stimulation or chemical stimulation). These receptors then pass the resulting impulses to the brain in the form of neural impulses.

Qualitative variations (sensory quality):

- Different sensory qualities are signaled by different neurons.
- Certain sensory qualities arise because of different patterns of activation across a whole set of neurons

Quantitative variations (intensity):

- Rates of neuron firing
- Total number of neurons triggered

Absolute threshold: the minimum intensity of stimulation that must occur before you experience a sensation.

Difference threshold:The smallest difference between two stimuli that you can notice.
(Weber's law:The just noticeable difference between two stimuli is based on a proportion of the original stimulus rather than on a fixed amount of difference.)
Signal detection theory: detecting a stimulus is a subjective decision with two components. Sensitivity to the stimulus in the presence of distractions from other stimuli and the criteria used to make the judgement from ambiguous information.

Sensory adaptation: the tendency to respond less to a stimulus that has been present and unchanging for some time.

Synesthesia: people with synesthesia experience cross-sensory experiences. They might taste a color for example.

**Vision**
Light (direct or reflected) can vary in intensity and wavelength.

Light first passes through the cornea (the eye's outer layer). The cornea focuses the incoming light, which then enters the lens. From there the light is bent farther inward and focused to form an image on the retina, the inner surface of the back of the eyeball which contains the sensory receptors. When the light hits the receptors (rods and cones) it is absorbed by pigments which convert light energy into electric energy. The electric signal is carried to the brain via ganglion cells.
Rods: are primarily used when there is little light, they are much more sensitive to light. No color vision.
Cones: are primarily used when there is more light. Color vision, there are three receptors; bleu/S (small wavelength), green/M (medium wavelength), and red/L (large wavelength).

Trichromatic theory: color vision arises as a result of the pattern of the activity across all three types of color cones.
Opponent-process theory: some colors are opposites of other colors (black-white, red-green, blue-yellow).
Both theories are correct.

Two systems:
- Where: path to parietal cortex, dorsal
- What: path to temporal cortex, ventral

The visual system is not a passive receiver of visual information but modulates, organizes, and interprets this information.

Color is categorized along three dimensions:
- Hue: consists of the distinctive characteristics that place a particular color in the spectrum. Depends primarily on the light's dominant wavelength.
- Saturation: the purity of the color. Depends on the mixture of wavelengths.
- Lightness: the color's perceived intensity. Depends on the total amount of light reaching the eye.

Gestalt perceptual grouping rules:
- Proximity: the closer two figures are to each other, the more likely we are to group thema and see them as part of the same object.
- Similarity: we tend to group  figures according to how closely they resemble each other.

Depth perception: we are able to perceive depth due to depth cues, these cues can be divided into two types:
- Binocular depth cues: are available from both eyes together and contribute to bottom-up processing.
- Monocular depth cues: are available from each eye alone and provide organizational information for top-down processing.

**<u>Topic 5: Learning</u>**
**Book: 6 Learning (207-244)**
Learning: a change in behavior resulting from experience.

We learn in three basic ways:
- Nonassociative learning: responding after repeated exposure to a single stimulus or event.
- Associative learning: linking two stimuli or events that occur together.
- Observational learning: acquiring or changing a behavior after exposure to another individual performing that behavior.

**Nonassociative learning**
The two most common forms are habituation and sensitization. For both types presynaptic neurons alter their neurotransmitter release. A reduction in neurotransmitter release leads to habituation. An increase in neurotransmitter release leads to sensitization.

Habituation: a decrease in behavioral response after repeated exposure to a stimulus. This allows a better ability to focus on what is new by ignoring which has already become familiar.
Dishabituation: an increase in a response because of a change in something familiar.

Sensitization: an increase in behavioral response after exposure to a stimulus.

**Associative learning: classical conditioning**
classical/pavlovian conditioning: A form of learning in which a neutral stimulus comes to elicit(veroorzaken) a response when it is associated with a stimulus that already produces that response.
Before training → training (Conditioning trials or reinforced trials) → after training (critical trials or unreinforced trials)

For learning to take place, the conditioned stimulus must come before the unconditioned stimulus.

Second-order conditioning: when a conditioned stimulus is paired with a new conditioned stimulus, the new stimulus produces a conditioned response.

Acquisition: the formation of an association between a conditioned stimulus and an unconditioned stimulus.
Extinction: the association between the conditioned stimulus and conditioned response can be eliminated by repeatedly presenting the conditioned stimulus alone.
Spontaneous recovery: a process in which a previously extinguished conditioned response reemerges after the presentation of the conditioned stimulus.

Stimulus generalization: the conditioned reaction can also be elicited by stimuli that are similar to the conditioned stimulus.
Stimulus discrimination: a differentiation between two simmilar stimuli when one of them is consistently associated with the undconditiond stimulus and the other not.

The conditioned stimulus functions as a sort of predictor of the unconditioned stimulus.
The conditioned response is a reaction in preparation of the unconditioned response.

**Associative learning: operant conditioning**
operant/instrumental conditioning: a form of learning in which the consequence of an action (response) determines the likelihood of that action being repeated.
Learned is the relationship between response and the consequence.

Law of effect: behavior is governed by its consequences. The tendency to give a response is strengthened if it’s followed by a reward and weakened if it’s not.

Reinforce: a stimulus that occurs after a response and increases the likelihood that the response will be repeated.
Shaping: consists of reinforcing behaviors that are increasingly similar to the desired behavior.

Primary reinforcer: stimulus that satisfies biological needs.
Secondary reinforcer: stimulus that serves as reinforcers but does not satisfy biological needs.

Positive reinforcement: increases the probability of that behavior being repeated.
Negative reinforcement: increases behavior through the removal of an unpleasant stimulus.

Contiguity: learning only occurs when the reinforcer or punishment immediately follows the response.
Contingency: the reinforcer or punishment should occur more likely after the response than otherwise.

blz 230-231

Behavior modification: the use of operant-conditioning techniques to eliminate unwanted behaviors and replace them with desirable ones.

**Observational learning**
Observational/social learning: the acquisition or modification of a behavior after exposure to another individual performing that behavior.

Modeling learning: copying the behavior of another individual.
Vicarious learning: learning the consequences of an action by seeing someone either be rewarded or punished after certain behavior.

Mirror neurons: neurons in the brain that are activated when one observes another individual engage in an action and when one performs a similar action.

**<u>Topic 6: Memory</u>**
**Book: 7 Memory (249-284)**
Perception: the organization of current stimuli.
Memory: the organization of stimuli from the past. The nervous system capacity to retain and retrieve skills and knowledge.

Memory operates over time in three phases:
- Encoding: the processing of information so that it can be stored.
- Storage and consolidation (in memory traces): the retention of encoded representation and the neural process by which encoded information becomes stored in the brain.
- Retrieval: the act of recalling or remembering stored information when it is needed.

Theory: once memories are activated, they need to be consolidated again to be stored back in memory.
reconsolidation: Neural processes involved when memories are recalled and then stored again for retrieval.

**Encoding**
Three memory systems: stage theory of memory.

- Sensory register: input from visuals (iconic memory) and hearing (echoic memory), all raw perceptual information.
- Short-term memory: Limited capacity (7, memory span), can expand it by recoding. Items disappear as a result of decay and replacement. By repeating stm > LTM
- Long-term memory: the storage of information. Nearly limitless capacity, long duration.

Now: memory is an active process requiring attention.
Short term memory is now working memory: information one is thinking about. An active processing unit that deals with multiple types of information.
long-term memory: information, stored during longer intervals, which is currently not actively used.

Serial position effect: the idea that the ability to recall items from a list depends on the order of presentation, with items presented early or late in the list remembered better than those in the middle.
Primacy effect: the better memory for items presented at the beginning of the list.
Recency effect: the better memory for the most recent items.
You rehearse the earliest items the most. As a result, that information is transferred into long-term memory. In contrast, the last few items are still in working memory.

encoding processes that promote later recall:
- Depth of processing: Shallow or deep processing.
- Understanding: a schema may help to understand and remember information.

Working memory can be separated from long-term memory. Still by most people, the two memory systems are highly interdependent.

Chunking: breaking down information into meaningful units. The more efficiently you chunk, the more you can remember.
Decisions on how to chunk information depend on schemas. Schemas are cognitive structures in long-term memory that help us perceive, organize, process, and use information. They provide structures for understanding events in the world. Schemas influence how we encode information in our daily lives.

Maintenance rehearsal: repeating the item over and over.
Elaborative rehearsal: encodes the information in more meaningful ways, such as thinking about the item conceptually or deciding whether it refers to oneself.
Semantic encoding activates more brain regions than shallow encoding and this greater brain activity is associated with better memory.

**Storage**
After encoding > storage in memory traces
Memory consolidation: the neural process by which encoded information becomes stored in memory. Transient and fragile potential memory contents are transformed into a more permanent state.
Long-term potentiation (LTP): strengthening of a synaptic connection, making the postsynaptic neurons more easily activated by presynaptic neurons.

Memories are stored in the specific brain regions engaged during the perception, processing and analysis of the material being learned.

After rehearsal > reconsolidation

After encoding > storage in memory traces

**Retrieval**
Memories are strengthened with retrieval, so one way to make durable memories is to practice retrieval.

A set of theories about memory organization is based on networks of associations. An item's features are linked so as to identify the item.
Each unit of information in the network is a node. Each node is connected to many other nodes. Nodes are simply bits of information. Activating one node increases the likelihood that closely associated nodes will also be activated.
Stimuli in working memory activate nodes in long-term memory. This activation increases the ease of access to that material and thus makes retrieval easier.

Retrieval cue: a cue that allows you to enter a memory trace. It can be anything that helps a person recall a memory.
Good cues:

- re-create the stage of mind in which the original learning occurred.
- A richer encoding context.

**Varieties of memory**
Long-term memory is composed of several interacting systems. These systems all share the same function, to retain and use information. But they encode and store different types of information in different ways.

Explicit memory (declarative): the system underlying conscious memories.
Episodic memory: Personally experienced events
Semantic memory: Facts and knowledge

Implicit memory (nondeclarative): The system underlying unconscious memories.
Procedural memory (motor memory): Involves motor skills, habits, and other behaviours used to achieve goals.
Classical conditioning: Associating two stimuli

**When memory fails**
Amnesia: a deficit in long-term memory, resulting from disease, brain injury, or psychological trauma, in which the individual loses the ability to retrieve vast quantities of information.
Anterograde amnesia: people lose the ability to form new memories.
Retrograde amnesia: people lose past memories.

Absentmindedness: inattentive or shallow encoding of events.

The major cause of this is failing to pay attention. Like when you are caught up in another activity

During most of our daily activities we are consciously aware of only a small portion of our thoughts and behaviors.

Memory decay: forgetting over time.
Most forgetting occurs because of interference from other information. Decay of memory traces with time. Caused by metabolic processes.
interference: after you sleep you forget less.
proactive interference: old information inhibits the ability to remember new information.
Retroactive interference: new information inhibits the ability to remember old information.

Blocking: when a person is temporarily unable to remember something.
It often occurs because of interference from things that are similar in some way.

Persistence: when unwanted memories are remembered in spite of the desire not tho have them.
Considerable research is under way to produce drugs that will erase unwanted memories.
But erasing memories leads to many ethical questions.

Memory bias: the changing of memories over time so that they become consistent with current beliefs or attitudes.
People often revise memories when they change attitudes and beliefs.

Flashbulb memories: vivid episodic memories for the circumstances in which people first learn of a surprising and consequential or emotionally arousing event.
Although normal forgetting takes place, people tend to repeat the same story over time.
What does not seem to change is the extent to which people feel confident in their memories.

Source misattribution: when people misremember the time, place, person, or circumstances involved with a memory.
Source amnesia: a form of misattribution that occurs when a person has a memory for an event but cannot remember where he or she encountered the information.
Childhood amnesia: the absence of early episodic memories.
Cryptomnesia: a person thinks they have come up with a new idea, but instead the person has retrieved an old idea from memory and failed to attribute the idea to its proper source,

suggestibility: the development of biased memories when provided with misleading information.
Different wordings of questions may alter memories.
Memories can be distorted or implanted with false information.
Memories can be altered by the intrusion of generic knowledge or semantic associations.
When a person imagines an event happening they form a mental image of the event. The person might later confuse that mental image with a real memory.

This all makes things like eyewitness testimony's troubling. Because how do you know if someone's story is accurate even when it is not, if the eyewitness themself really believe it is exactly what happened?

**<u>Topic 7: Thinking and language</u>**
**Book: 8 Thinking, Language, and Intelligence (289-319)**

**Mental representations**
Mental representations are components of knowledge:
- Analogical representation: mental representations that have some of the physical characteristics of objects.
- Symbolic representation: abstract mental representations that do not correspond to the physical features of objects or ideas.

Mental maps rely on both analogical and symbolic representations.

A concept is a category or class of related items. It consists of mental representations of those items. A concept ensures that we do not have to store every instance of an object individually.
Two leading models of forming concepts:
- Prototype model: within each category, there is a best example, a prototype, for that category. Each member of a category varies in how much it matches the prototype.
- Exemplar model:all members of a category are examples. Together they form the concept and determine category membership.

We can use schema's for two reasons:
- common situations have consistent rules
- People have specific roles within a situational context.

But schema's can sometimes have unintended consequences, like sexism or racism or other stereotypical beliefs.
script: a schema that directs behavior over time within a situation.
The schemas and scripts that children learn are likely to affect their behavior when they are older.

**Decision making**
Decision making: an attempt to select the best option out of alternatives options.

Normative decision theories: define how people should make decisions.
Expected utility theory: decisions should be made to maximize the outcome. This should depend on the values of the alternative options and the probabilities associated with the alternative options.

Descriptive decision theories: indicate how people actually make choices.

Heuristics: shortcut used to reduce the amount of thinking that is needed to make decisions. This often occurs unconsciously. It results in deciding quickly.
But it can result in errors or faulty decisions:
- Anchoring: the tendency to rely on the first piece of information encountered or information that comes most quickly to mind.

- Framing: the tendency to emphasize the potential losses or potential gains from at least one alternative.
- Availability heuristic: making a decision based on the answer that most easily comes to mind.
- Representativeness heuristic: making a decision based on how typical an individual, object or event belongs in a certain category.
- Affective forecasting heuristic: the tendency for people to overestimate how events will make them feel in the future.
- The paradox of choice: when too many options are available, people experience conflict and indecision.

A good strategy is to statisifice rather than optimize your decisions.

**Problem solving**

Problem solving: the thinking process in which we overcome obstacles to move from an initial state to a goal state.

Obstacles in problem solving:

- complexity
- Mental set:
    - Instruction or practice
    - Perceptual organization of the problem
    - Functional fixedness: having fixed ideas about the typical functions of objects.

Overcoming obstacles in problem solving:

- Understanding: formulate subgoals
- Restructuring
- Use an algorithm
- Use an analogy

**Language**

Language: a system of communication using sounds and symbols according to grammatical rules.
Syntax: rules
Semantics: meanings

As the brain develops, so does the ability to speak and form sentences. Development occurs in an orderly way.
Listening preferences in newborns are influenced by the language(s) spoken by mothers during pregnancy. Newborns can discriminate all the phonemes but lose this ability after 6 months.

Aphasia: a language disorder that results in deficits in language comprehension and production.
Expressive aphasia (Broca's aphasia): interruption of ability to speak.
Receptive aphasia (Wernicke's aphasia): difficulties to understand the meaning of words.

**<u>Topic 8: Intelligence</u>**

**Book: 8 Thinking, language, and intelligence (319-332)**

Intelligence: the ability to use knowledge to reason, make decisions, make sense of events, solve problems, understand complex ideas, learn quickly, and adapt to environmental challenges.

General psychology: there is a focus on general psychological rules that applicable to all people
Differential psychology: focuses on differences between individuals

**Intelligence testing**
You can measure intelligence:
IQ = MA / CA * 100
MA = Mental age
CA = Chronological age

The IQs of adults are measured differently. It is measured in comparison with the average adult and not with adults at different ages.

Intelligence tests have high reliability and high predictive validity.

**The psychometric approach**
The psychometric approach is an attempt to understand the nature of intelligence by studying the pattern of results obtained on intelligence tests.

General intelligence:
- Linguistic ability
- Numerical ability
- Spatial ability

Raymond Cattell:
- Fluid intelligence: being able to understand abstract relationships and think logically without prior knowledge. It declines steadily throughout the adult years.
- Crystallized intelligence: knowledge acquired through experience, and the ability to use this knowledge to solve problems. It grows steadily throughout the adult years.

Other researchers like Howard Gardner:
- Analytical intelligence: being good at problem solving, completing analogies, figuring out puzzles, and other academic challenges.
- Creative intelligence: the ability to gain insight and solve novel problems.
- Practical intelligence: everyday task.

Emotional intelligence: a form of social intelligence that emphasizes the abilities to manage, recognize, and understand emotions and use emotions to guide appropriate thought and action.
But is EI really a type of intelligence or not?

**Intelligence beyond the IQ test**
Standard intelligence tests typically fail to capture other types of intelligence.

Sternberg: You have analytic, creative, and practical intelligence

Salovey & Mayer: emotional intelligence
Gardner: multiple

**The relation between IQ and cognitive performance**
Different measures correlate with g.

- Simple RT and choice RT task
- Discrimination task performance
- Working memory capacity
- Executive control

General intelligence scores are closely related to how people process information in working memory. The link between these two may be attention.

**The roots of intelligence**
The score on an intelligence test is affected by both genes and the environment.

Impoverished environments: correlate negative with IQ, they have an big influence
Enriched environments: correlate with positive IQ

Heritability:
H = Genetic variance / Phenotypic variance

Stereotype threat: the effect of the apprehension or fear that some people might experience if they believe that their performances on tests might confirm negative stereotypes about their racial group.
It causes distraction and anxiety, interfering with performance by reducing the capacity of short-term memory and undermining confidence and motivation.

**<u>Topic 9: scientific reasoning</u>**
**Book: 1 Psychology is a way of thinking (5-19), 2 Sources of information: Why research is best and how to find it (22-38)**
The law of attraction: the belief that by focusing on positive or negative thoughts people can bring positive or negative experiences in their lives. “like attracts like”
It is presented as a law of nature that is science based and that if you don’t believe in it then the universe realizes that it doesn’t work for you.

**Fixation of belief**
Belief: an attitude of some person toward a particular claim.
There is no direct connection between belief and truth.

Methods to fix beliefs:

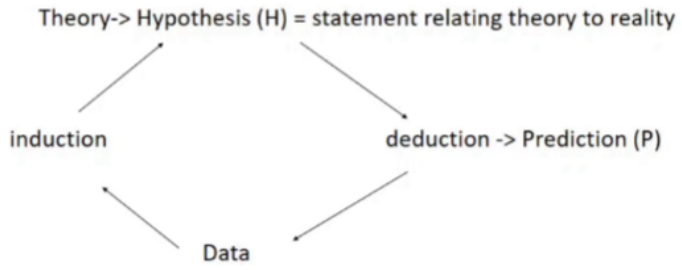
- Personal experience
  Basing conclusions on personal experiences is problematic because daily life usually doesn’t include comparison experiences.
  A comparison group enables us to compare what would happen both with and without the thing we are interested in.
  present bias: our failure to consider appropriate comparison groups.

- Intuition
  Accepting conclusions just because it makes sense or feels natural
  Availability heuristic: things that pop up easily in our mind tend to guide our thinking. This leads us to overestimate events.
- Authority
  Confirmation bias: the tendency to look only at information that agrees with what we want to believe.
  Before taking the advice of authorities, ask yourself about the source of their ideas. Because authorities can also base their advice on their own experience or intuition. And even if it is based on a research, not all research is equally reliable.

The scientific method fixes belief on the basis of empirical research. It combines proper reasoning and systematic empiricism to reach conclusions.

Basing conclusions on personal experiences is problematic because daily life usually doesn't include comparison experiences.
A comparison group enables us to compare what would happen both with and without the thing we are interested in.

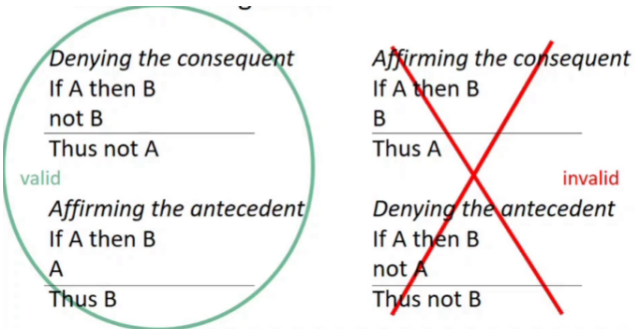
**What is science?**
Empiricism: involves using evidence from the senses or from instruments that assist the senses as the basis for conclusions.
Empirical cycle (theory-data cycle)




Induction: specifying a general rule based on specific observations. The conclusion is often plausible but not inescapable.
Deductive: specifying a conclusion about something specific on the basis of a general rule. The conclusion inevitably follows from the assumptions/premises.

Scientific method: a method to provide good justification for the belief that a claim or statement is true or false.
Goal of science: to find good justification for the belief that a claim or statement is true or false. You do that by using a conditional argument.

**The logic behind science.**
Conditional argument: is based on a conditional statement.




The type of argument used to justify or falsify a scientific hypothesis is the same across different scientific disciplines.
The conclusion of the argument will be either the hypothesis or its negation.

Confound: when you think one thing caused an outcome but in fact other things changed too, so you are confused about what the cause really was.

**Scientific reasoning**
Hypothesis + initial conditions + auxiliary assumptions → prediction
If [H and IC and AA] then P
If [not-H and IC and AA] then probably not-p

research is usually more accurate than individual experience, sometimes personal stories contradict the research results. The research is then not necessarily wrong. The results of behavioral research are probabilistic, its findings do not explain all cases all of the time.

Studies don't prove anything. A study's data supports or is consistent with a theory.

**Falsifiability criterion**
Falsifiable: a feature of a scientific theory in which it is possible to collect data that will indicate that the theory is wrong.

To find good justification for the belief that a hypothesis is true, the hypothesis needs to be potentially falsifiable.
Scientific hypotheses need to generate predictions that might not be true, the falsifiability criterion.

**Theories**
Theory: a set of statements that describes general principles about how variables relate to one another.

Evolution theory: life on earth evolved gradually by natural selection (darwin).

Intelligent Design Theory: life on earth is created by an intelligent designer.

Mother-infant attachment

- The cupboard theory: a mother is valuable to a baby mammal because she is a source of food. Over time the sight of the mother acquires positive value because she is the “cupboard” from which food comes.
- Contact comfort theory: babies are attached to their mothers because of the comfort of ther warm, fuzzy fur.

**<u>Topic 10: Validity</u>**
**Book: 3 Three claims, four validities: Interrogation tools for consumers of research (55-81), 11 More on experiments: Confounding and obscuring variables (322-341)**

**Basic knowledge**
Measured variable: a variable whose levels are simply observed and recorded.
Manipulated variable: a variable a researcher controls, usually by assigning study participants to different levels of that variable.

Independent variables: manipulated variables, the variables you are interested in.
Dependent variables: measured variables.
Control variables (constant): variables that are being held constant on purpose.
Confounding variables: variables that change simultaneously with the independent variables.
Conceptual variable (construct): a variable stated at an abstract (theoretical) level.
Operational variable (operational definition): the specific way in which a construct is manipulated or measured in a study.

The goal of science is to find good justification for the belief that a specific claim or statement (hypothesis) is true or false.

Three type of claims:
- Frequency claims: focus on one variable.
  Claims that mention the percentage of a variable, the number of people who engage in some activity, or a certain group’s level on a variable can all be called frequency claims.
- Association claims: focus on the relationship between at least two variables.
  The variables are measured and the relationship between them is tested, correlational study
- Causal claim: focus on causation: a change in one variable is responsible for changing the value of another variable. So it suggests that one variable comes first in time and acts on the other variable.
  A study here has to establish that the two variables are correlated, it must show that the causal variable came first and the outcome variable came later(temporal precedence), and it must establish that no other explanations exist for the relationship(internal validity).

A valid claim is reasonable, accurate, and justifiable.

To operationalize a concept of interest means to turn it into a measured or manipulated variable.

**Construct validity**

Construct validity: an indication of how well a conceptual variable is measured or manipulated in a study.

Threats to construct validity:

- Inadequate operational definition
- Mono-operation bias

**External validity**

External validity: an indication of how well the results of a study generalize to, or represent, individuals, settings, places, and times besides those in the study itself.
Is it possible to generalize?

Threats to external validity:

- Selection biases
- Study setting is different from other settings
- Specific time at which the study is performed is different from other times

**Internal validity**

Internal validity: the extent to which it is possible to rule out alternative explanations for a casual relationship between two variables
Can we rule out alternative explanations?

Threats to internal validity:

- Selection effects: the different independent variable groups have systematically different types of participants.
- Design confounds
- Order effects: we do not know if the independent variable is really having an effect, or whether the participants are just getting tired, bored, or well-practiced. (practice effects, carry over effects)
- Maturation: a change in behavior that emerges more or less spontaneously over time.
- History: a “historical” or external factor that systematically affects most members of the treatment group at the same time as the treatment itself, making it unclear whether the change is caused by the treatment received.
- Regression to the mean: when a group average (the mean) is unusually extreme at time 1, the next time that group is measured, it is likely to be less extreme/closer to the typical or average performance.
- Attrition/experimental mortality: a reduction in participant numbers that occurs when people drop out before the end.
- Testing: a change in the participants as a result of taking a test more than once.
- Instrumentation: when a measuring instrument changes over time.
- Observer bias: when researchers' expectations influence their interpretation of the results.
- Demand characteristics: when participants guess what the study is supposed to be about and change their behavior in the expected direction.
- Placebo effects: when people receive a treatment and really improve, but only because they believe they are receiving a valid treatment.

**Statistical validity**
Statistical validity: the extent to which a study's statistical conclusions are accurate and reasonable.
How well do the numbers support the claims?

Threats to statistical validity:
- Violated assumptions of the test statistics
- Fishing and error rate problem: seeing things that aren't there
- Low statistical power

Which of the validities is the most important? → Depending on their goals, sometimes researchers place a lower priority on certain validities. You should not necessarily conclude that a study is faulty just because it did not meet all the validities.

**<u>Topic 11: Measurement, reliability, and descriptive research</u>**
**Book: 5 Identifying good measurement (116-133), 6 Surveys and observations: Describing what people do (153-174), 7 Sampling: Estimating the frequency of behaviors and beliefs (178-186)**
Conceptual definition (construct): the researcher's definition of the variable in question at a theoretical level.
Operational definition: represents a researcher's specific decision about how to measure or manipulate the conceptual variable.

**Measurement scales**
Measurement: a systematic way to assign numbers or names to objects and their features

Operational/behavioral measure: operationalizes a variable by recording observable behaviors or physical traces of behaviors.
Physiological measure: operationalizes a variable by recording biological data.

Categorical variables: are categories. A researcher might decide to assign numbers to the levels of a categorical variable, the numbers do not have numerical meaning.
Quantitative variables (continuous variables): are coded with meaningful numbers.

Measurement scales:
- Nominal scale: differences, categorical variables.
- Ordinal scale: ranked order, quantitative variables.
- Interval scale: equal intervals (add/subtract), quantitative variables.
- Ratio scale: absolute zero (ratio's), quantitative variables.

Importance of measurement scales:
- Determines the possible statistical operations
- Determines the possible conclusions

**Reliability**
Observed score = true score + error score
Error score: the difference between the measured value and the real value which is affected by the test (situation) and the person.

Different types of reliability:

- Test reliability
    - Test-retest reliability
    - Internal reliability: a participant gives a consistent pattern of answers, no matter how the researchers phrase the question.
- Interrater reliability: consistent scores are obtained no matter who measures the variable.
- Experimental reliability

**Descriptive research**

Observational research: the process of watching people or animals and systematically recording how they behave or what they are doing.
You can use this to describe nature and to generate hypotheses.
Observational research can be the basis for frequency claims.

Problems observational research:

- people are not always good at oveservation: this is a threat to the interrater reliability
- Low internal validity
- Threats to construct validity: observer bias, observer effects, reactivity

Survey research: a method of posing questions to people on the telephone, in personal interviews, on written questionnaires, or via the internet.

Problems survey research:

- Low internal validity
- Threats to construct validity
    - Question wording: leading questions, double-barreled questions, and negatively worded questions
    - Question order
    - The use of response sets: a shortcut people can take when answering. People might adopt a consistent way of answering all the questions. (acquiescence/yes-saying or fence sitting)
    - Trying to look good or bad: socially desirable responding/faking good/faking bad.
    - People are not always capable of reporting accurately about their feelings, thoughts, and behavior

Descriptive research results in frequency claims

Population: the entire set of people or products in which you are interested.
Sample: a smaller set, taken from that population.
Census: when you test the whole population.

Threats to external validity:

- Biased or unrepresentative sample
    - Conveniences: studies may use samples consisting of those who are easy to contact

    - Self-reflection: many studies use samples consisting of only those who volunteer

To obtain representative sample there are various techniques:

- Random sample
- Different rewards
- Replication

Threats to statistical validity:

- A too small sample size

**<u>Topic 12: Correlations and experiments</u>**

**Book: 8 Bivariate correlational research (203-227), 9 Multivariate correlational research (241-248), 10 Introduction to simple experiments (277-311), 11 More on experiments: Confounding and obscuring variables (342-357).**

**Correlational research**

Correlational research leads to association claims.
When both variables in an association are measured on quantitative scales, a scatter plot is usually the best way to represent data.
The mean (the average) for all the cases is represented in a bar graph.

Correlation coefficient (r): a measure indicating the degree of linear relation between two variables.
It usually varies between -1.00 and 1.00. The absolute value indicates the strength of the relation. The sign indicates the direction of the relation.

Statistical validity: does the data support the claim?

- Effect size (r): what is the strength of the correlation? It can indicate the importance of a result.
- Could outliers be affectinig the correlation? An outlier is an extreme score that stands out from the pack. They matter the most when a sample is small.
- How precise is the estimate? To communicate the precision of the estimate of r, researchers report a confidence interval (CI). When an estimate is based on a small sam;le, it is less stable.
- Has it been replicated? Replication is conducting a study again.
- Is there a restriction of range? If there is not a full range of scores on one of the variables in the association, it can make the correlation appear smaller than it really is.
- Is the association curvilinear? A curvilinear association is when a relationship between two variables is not a straight line. When the slope of the scatterplot goes up and then down, r does not describe the pattern well.

The presence of a correlation indicates an association but does not indicate causality!

- Directionality problem: the direction of causation is unknown.
  Sometimes researchers attempt to establish temporal precedence on the basis of correlational research, by using the cross-lagged-panel procedure.
- 3rd variable problem: a 3rd variable causes both of the things you think have a correlation.

Sometimes researchers attempt to establish internal validity on the basis of correlational research, by separating the participants into subgroups on the basis of a reasonable 3rd variable.

Criteria required to establish causation:

- Covariance of cause and effect: an association between a causing variable and an effect variable.
- Temporal precedence (directionality problem): the causing variable must come before the effect variable.
- Internal validity (third-variable problem):there shouldn't be an alternative third variable.

Spurious association: the correlation is there, but only because of some third variable.
Moderator: a variable that changes the relationship between two variables.

External validity: Can the association be generalized?:
The size of the sample does not matter as much as the way the sample was selected from the population of interest.
When a correlational study didn't use a random sample you should not automatically reject the association for that reason. Many associations do generalize, even to samples that are very different from the original one.

What if the correlation is zero?

- Restriction of range (or truncated range)
- Non-linear relationship
- No association

Multivariate designs: a study designed to test an association involving more than two measured variables.
Longitudinal design: can provide evidence for temporal precedence by measuring the same variables in the same people at several points in time.

Auto correlations: the correlation of a variable with itself, measured at two different times.
Cross-sectional correlations: when two variables measured at the same point in time are correlated. This can not establish temporal precedence.
Cross-lag correlations: whether the earlier measure of one variable is associated with the later measure of the other variable. This establishes temporal precedence.

**Experimental research**
An experiment occurs when a particular comparison is produced while other aspects of the situation are held constant.

Manipulated variable: a variable that is controlled.
Measured variables: take the form of records of behavior or attitudes.
Independent variable: the manipulated variable.
Dependent variable: the measured variable
Control variable: any variable that an experimenter holds constant on purpose.

In a graph the independent variable is on the x-axis and the dependent variable on the y-axis.

Confound: a potential alternative explanation for a research finding.
Design confound: when a second variable happens to vary systematically along with the intended independent variables.

Null result:

- The independent variable did not affect the dependent variable.
- Not enough between-groups difference
    - Manipulation was too weak
    - Insensitive measures
    - Ceiling effect:all the scores are squeezed together at the high end.
    - Floor effect: all the scores are squeezed together at the low end.
- Too much between-groups differences
  Noise: there is too much unsystematic variability within each group.
    - Individual differences
    - Unreliable measurements
- Not enough power (small n): the likelihood that a study will return an accurate result when the independent variable really has an effect.

Manipulation checks: separate dependent variables that experimenters include in a study specifically to make sure the manipulation worked. Help detect weak manipulations, ceilings and floors.

How to assign subjects to the different levels of the independent variable(s)?

- Between-subject design (independent-groups design): separate groups of participants are placed into different levels of the independent variable.
- Within-subjects design (within-groups / repeated-measures design): each person is presented with all levels of the independent variable. Like the posttest-only design and the pretest/posttest design.

Repeated-measure design: a type of within-groups design in which participants are measured on a dependent variable more than once, after exposure to each level of the independent variable.
Concurrent-measure design: participants are exposed to all the levels of an independent variable at roughly the same time, and a single attitudinal or behavioral preferences is the dependent variable.

Counterbalancing: in repeated measure experiments, presenting the levels of the independent variable to participants in different sequences to control for order effects.
Full counterbalancing: all possible condition orders are represented. Each group of participants could be randomly assigned to one of all the sequences.
Partial counterbalancing: only some of the possible condition orders are represented. You can use the latin square for this.
Latin square: a formal system to ensure that every condition appears in each position at least once. The first row is set up, then the conditions simply go in order down each column.

**<u>Topic 13: Single-factor and factorial designs</u>**
**Book: 12 Experiments with more than one independent variable (363-394)**

**Single-factor designs**
Between-subjects design
Advantage: there is no chance that one level of the independent variable affects performance under the other level of the independent variable.
Disadvantage: there is a chance that groups differ.
You use between-subject design when the independent variable has a permanent effect, and when there is a large risk of order effects.

Within-subject design.
Advantage: there is no chance of group differences, each subject serves as his own control.
Disadvantage: there is a chance of order effects.
You use within-subjects design when the independent variable does not have a permanent effect, and when order effects can be avoided by counterbalancing or random order.

Within-subject designs are more efficient than between-subject designs (more power).

**Factorial designs**
Adding an additional independent variable allows researchers to look for an interaction effect.
Interaction effect: whether the effect of the original independent variable depends on the level of another variable.

Factorial design: a design in which there are two or more independent variables. Each level of every independent variable occurs with all levels of the other independent variable.
In a factorial design with two independent variables, the first two results obtained are the main effects for each independent variable. The third result is the interaction effect, the difference in differences.
Interactions are more important than the main effects.

When results from a factorial design are plotted as a line graph and the lines are not parallel, there may be an interaction.
You can also estimate interactions from a bar graph. You need to imagine drawing a line to connect the tops of each pair of same-color bars. Then look of these are nearly parallel or not.

Participant variable: a variable whose levels are selected, not manipulated.

Between-subjects design: A and B are both between-subjects factors.
Within-subjects design: A and B are both within-subjects factors.
Mixed design: one of the factors is a between-subjects design factor, the other is a within-subjects factor.

Advantages:
- Efficiency
- Generalization

- Information on interaction between factors

Crossover interaction: the lines in a graph of the interaction cross each other.

**<u>Topic 14: Small-N and quasi-experimental design</u>**
**Book: 13 Quasi-experiments and small-N designs**

**Small-N studies**
Small-N studies are typically used when there are only a few individual differences.
Two characteristics are the large external control and the researcher performs many observations.
So they obtain a lot of information from just a few cases.

Small-N studies:
- Psychophysics "if you've seen one, you've seen them all"
- Behavior-change studies
    - Stable baseline design (AB design): a practitioner or researcher observes behavior for an extended baseline period before beginning a treatment or other intervention.
      A = behavior during baseline, B = behavior during or after treatment.
    - Multiple-baseline design:researchers tagger their introduction of an intervention across a variety of individuals, times, or situations to rule out alternative explanations.
    - Reversal design (ABA or ABAB design): researchers observe a problem behavior both with and without treatment, but then take the treatment away for a while to see whether the problem behavior returns. They subsequently reintroduce the treatment to see if the behavior improves again.
- Case studies: tells us what can happen, they don't tell us what typically happens.

Small-N studies often aim to generate causal claims. But the internal validity is often not quite high. It is important to question how well each variable was manipulated and measured, how well the results generalize, and how well the numbers support the conclusions.

Single-N design: study on one person.

**Quasi-experimental designs**
With a quasi-experimental design the experimenter selects rather than manipulates the independent variables. This can be environmental or subject variables.
So the researchers do not have full experimental control.

Quasi-experimental designs:
- Pretest/posttest design (AB design)
- Nonequivalent control group posttest design: the participants are not randomly assigned to groups and are tested only once, after exposure to one level of the independent variable or the other.

- Nonequivalent control group pretest/posttest design: Participants are not randomly assigned to groups and are tested both before and after some intervention.
  participants → measure → treatment/no treatment → measure
- Interrupted time-series design: multiple measurements before treatment and multiple measurements after treatment.
- Nonequivalent control group interrupted time-series design: participants are not randomly assigned and there are multiple measurements before interrupting and multiple measurements after treatment. The presence or timing of the interrupting event differs among the groups.
  participants → measures → treatment/no treatment → measures

Quasi-experimental studies often aim to generate causal claims. But the internal validity is not as high as in a true experiment. It is important to question how well each variable was manipulated and measured, how well the results generalize, and how well the numbers support the conclusions.

Wait-list design: all the participants plan to receive treatment but are randomly assigned to do so at different times.